EXPERIMENT NO. 01

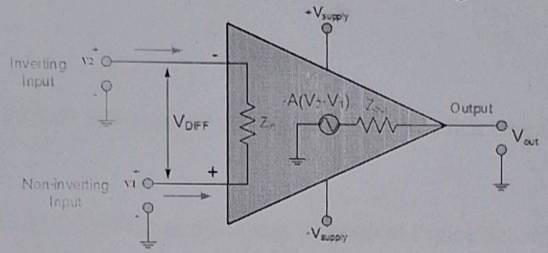
AIM: Simulation of inverting amplifier circuit using PSPICE

PROBLEM STATEMENT: Design inverting op-amp and measure the gain and plot the input and output waveforms.

SOFTWARE REQUIRED: PSPICE software

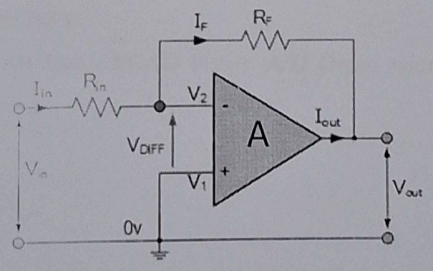
THEORY:

Operational Amplifier commonly known as Op-Amp, is a linear electronic device having three terminals, two high impedance input and one output terminal. Op-Amp can perform multiple function when attached to different feedback combinations like resistive, capacitive or both. Generally it is used as voltage amplifier and the output voltage of the Op-Amp is the difference between the voltages at its two input terminals. Op-Amp shows some properties that make it an ideal amplifier, its open loop gain and input impedance is infinite (i.e. Practically very high), Output impedance and offset voltage is zero(i.e. Practically very low) and bandwidth is infinite(i.e. Practically limited to frequency where its gain become unity).



Inverting Op-Amp:

The open loop gain (Ao) of the Op-Amp is very high which makes it very unstable, so to make it stable with a controllable gain, a feedback is applied through some external resistor (Rf) from its output to inverting input terminal(i.e. Also known as negative feedback) resulting in reduced gain(closed loop gain, Av). So the voltage at inverting terminal is now the sum of the actual input and feedback voltages, and to separate both a input resistor (Ri) is introduced in the circuit. The non-inverting terminal of the op-amp is grounded, and the inverting terminal behaves like a virtual ground as the junction of the input and feedback signal are at the same potential.



The close loop gain (A_{cl}) is given by:-

$$A_{cl} = \frac{V_{out}}{V_{in}} = -\frac{R_F}{R_{in}}$$

CIRCUIT DIAGRAM:

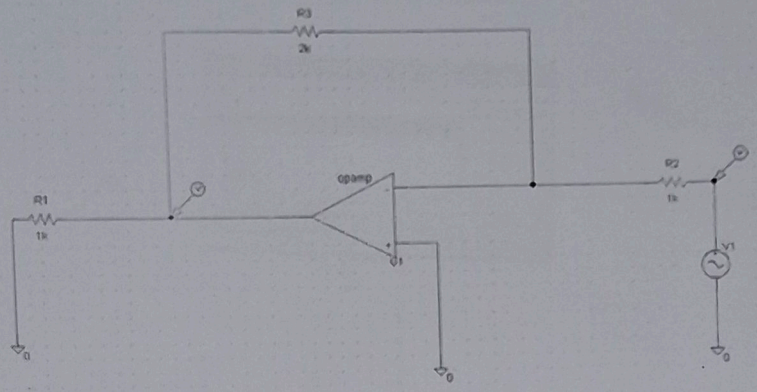


FIGURE 1: Schematic of **INVERTING AMPLIFIER** Circuit

PROCEDURE:

1. Design INVERTING AMPLIFIER as per circuit diagram in Pspice Schematic.
2. Use Current Marker for Input and Output waveform.
3. Select Transient Analysis as per fig. no 2.
4. Simulate the circuit and study the Input/output waveforms.
5. Most commonly error occurred during simulation is Netlist/ERC error, it is due to incomplete connections, you will observe this error as Floating point error, open Pspice message viewer double click on the error line, then it will revert back to schematic window and mouse cursor indicate the location of error in the schematic.
6. Once the schematic is error free, OrCAD Pspice A/D Demo window will open and following waveform is observed.

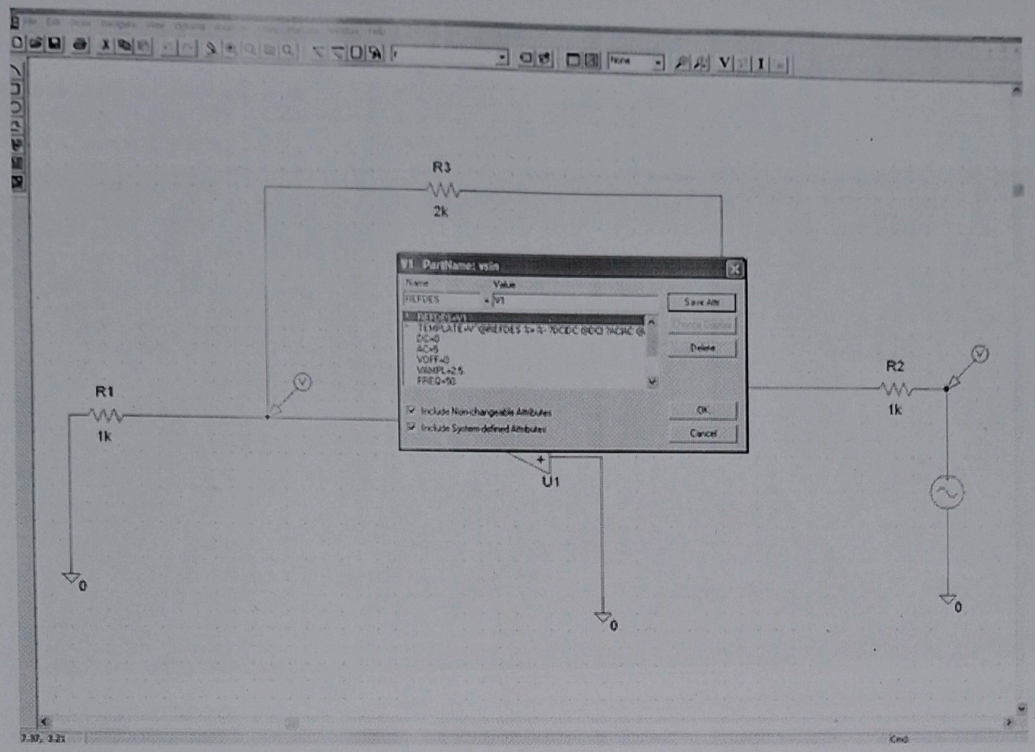


FIGURE 2(a): Setup Analysis

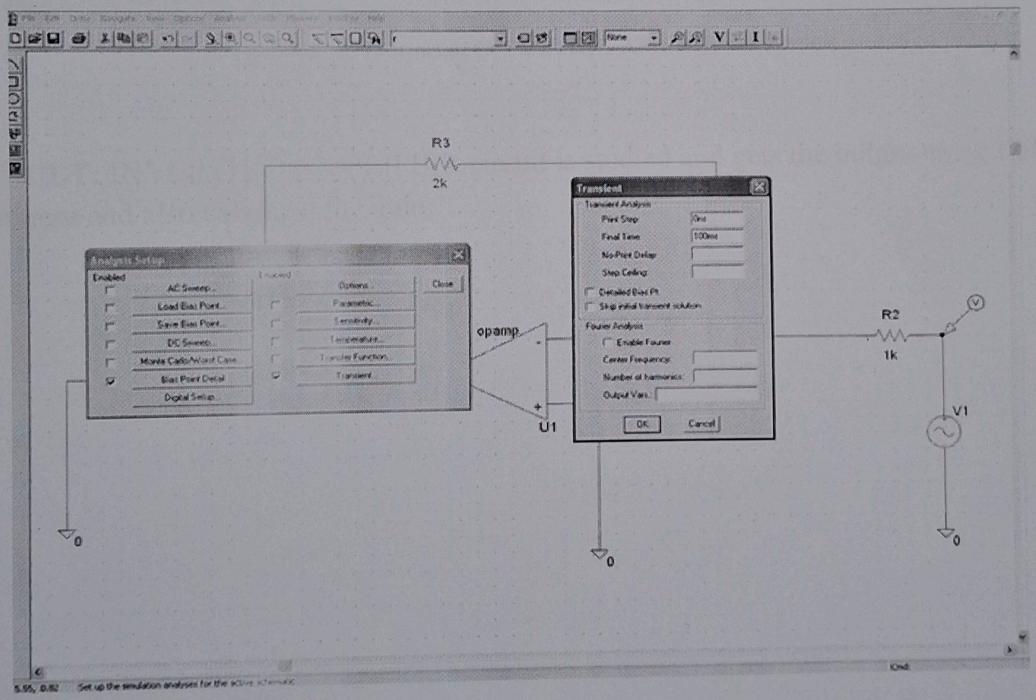


FIGURE 2(b): Setup Analysis

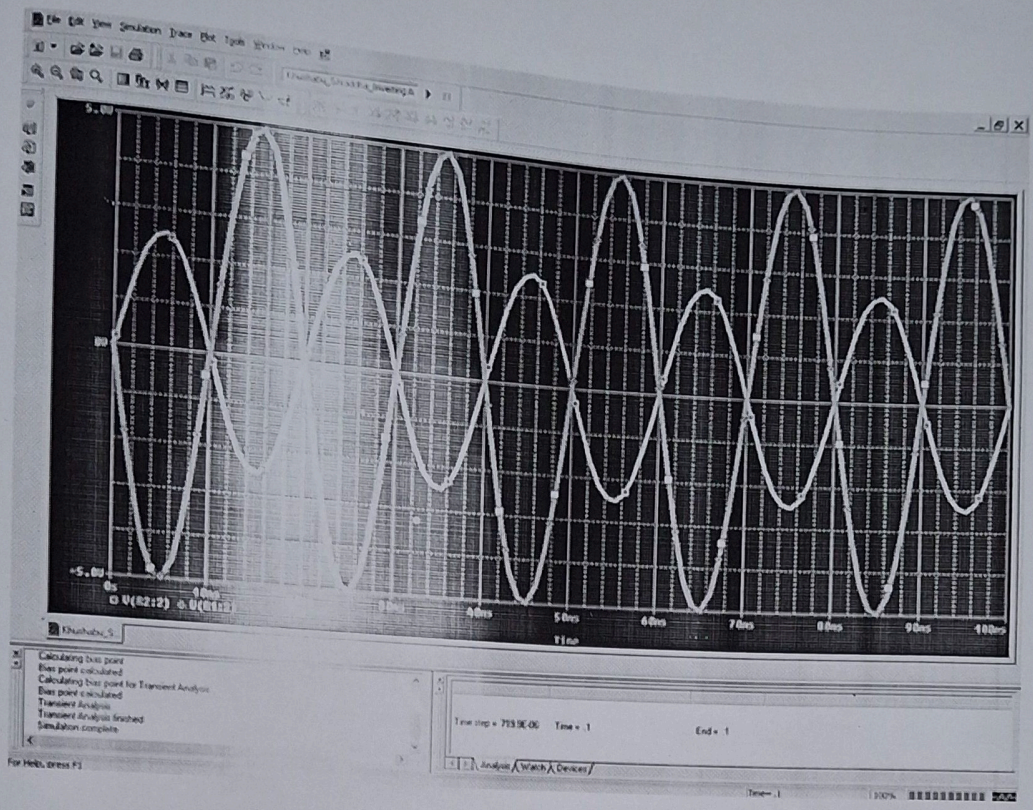


FIGURE 3: Output of **INVERTING AMPLIFIER Circuit**

RESULT: INVERTING AMPLIFIER circuit is studied and gets the output using PSPICE Software and also calculate the gain.